# General Relativity

## Final Examination — Model Solutions

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## Question 1. *(2 points)*

**Transformation of the Riemann tensor:**

Let $x^\mu$ and $x^{\mu'}$ be two coordinate systems related by a smooth, invertible coordinate transformation $x^\mu \mapsto x^{\mu'}(x)$. The Riemann curvature tensor ${R^\mu}_{\nu\rho\sigma}$ is a tensor of type $(1,3)$ (one contravariant index and three covariant indices). Under a change of coordinates $x^\mu \mapsto x^{\mu'}$, its components transform according to the tensor transformation law:

$$ {R^{\mu'}}_{\nu'\rho'\sigma'}(x') = \frac{\partial x^{\mu'}}{\partial x^\mu}\frac{\partial x^\nu}{\partial x^{\nu'}}\frac{\partial x^\rho}{\partial x^{\rho'}}\frac{\partial x^\sigma}{\partial x^{\sigma'}}{R^\mu}_{\nu\rho\sigma}(x)\,. \tag{1} $$

**Transformation of the Ricci tensor:**

The Ricci tensor is defined in any coordinate system by contracting the first (contravariant) index with the third (covariant) index of the Riemann tensor:

$$ R_{\mu\nu} = {R^\rho}_{\mu\rho\nu}\,. \tag{2} $$

In the primed coordinate system $x^{\mu'}$, the components of the Ricci tensor are therefore given by $R_{\mu'\nu'} = {R^{\rho'}}_{\mu'\rho'\nu'}$. Setting $\mu' = \rho'$ in equation (1) and summing over the repeated index $\rho'$ (with dummy indices relabelled appropriately to avoid confusion), we obtain:

$$ R_{\mu'\nu'}(x') = {R^{\rho'}}_{\mu'\rho'\nu'}(x') = \frac{\partial x^{\rho'}}{\partial x^\alpha}\frac{\partial x^\mu}{\partial x^{\mu'}}\frac{\partial x^\beta}{\partial x^{\rho'}}\frac{\partial x^\nu}{\partial x^{\nu'}}{R^\alpha}_{\mu\beta\nu}(x)\,. \tag{3} $$

By the multivariable chain rule, the Jacobian matrices $\frac{\partial x^{\rho'}}{\partial x^\alpha}$ and $\frac{\partial x^\beta}{\partial x^{\rho'}}$ are inverses of one another:

$$ \frac{\partial x^{\rho'}}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x^{\rho'}} = \frac{\partial x^\beta}{\partial x^\alpha} = \delta^\beta_\alpha\,, \tag{4} $$

where $\delta^\beta_\alpha$ is the Kronecker delta. Substituting this identity into the expression for $R_{\mu'\nu'}(x')$ yields:

$$ \begin{aligned} R_{\mu'\nu'}(x') &= \delta^\beta_\alpha\frac{\partial x^\mu}{\partial x^{\mu'}}\frac{\partial x^\nu}{\partial x^{\nu'}}{R^\alpha}_{\mu\beta\nu}(x) \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}}\frac{\partial x^\nu}{\partial x^{\nu'}}{R^\alpha}_{\mu\alpha\nu}(x) \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}}\frac{\partial x^\nu}{\partial x^{\nu'}}R_{\mu\nu}(x)\,. \end{aligned} \tag{5} $$

This is precisely the transformation law for the components of a $(0,2)$ tensor under a change of coordinates. Hence $R_{\mu\nu}$ transforms as a rank $(0,2)$ tensor.

## Question 2. *(3 points)*

We are given a manifold equipped with a metric $g_{\mu\nu}$ and a totally antisymmetric $(0,3)$ tensor $H_{\mu\nu\rho} = H_{[\mu\nu\rho]}$, along with the connection

$$\hat{\Gamma}_{\mu\nu}{}^{\rho} = \Gamma_{\mu\nu}{}^{\rho} + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \,, \tag{6}$$

where $\Gamma_{\mu\nu}{}^{\rho}$ are the components of the Levi-Civita connection.

a) **Torsion and metric compatibility:**

*1. Is $\hat{\Gamma}_{\mu\nu}{}^{\rho}$ torsion-free?*

The torsion tensor $\hat{T}_{\mu\nu}{}^{\rho}$ of a connection $\hat{\Gamma}_{\mu\nu}{}^{\rho}$ is defined as twice the antisymmetric part of the connection coefficients in their lower indices:

$$\hat{T}_{\mu\nu}{}^{\rho} = 2\hat{\Gamma}_{[\mu\nu]}{}^{\rho} = \hat{\Gamma}_{\mu\nu}{}^{\rho} - \hat{\Gamma}_{\nu\mu}{}^{\rho} \,. \tag{7}$$

Substituting equation (6):

$$\hat{T}_{\mu\nu}{}^{\rho} = \left(\Gamma_{\mu\nu}{}^{\rho} - \Gamma_{\nu\mu}{}^{\rho}\right) + \frac{1}{12} g^{\rho\sigma} \left(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}\right) \,. \tag{8}$$

Since the Levi-Civita connection $\Gamma_{\mu\nu}{}^{\rho}$ is torsion-free, it is symmetric in its lower indices, $\Gamma_{\mu\nu}{}^{\rho} = \Gamma_{\nu\mu}{}^{\rho}$, so the first term vanishes. Since $H_{\mu\nu\sigma}$ is totally antisymmetric, it satisfies $H_{\nu\mu\sigma} = -H_{\mu\nu\sigma}$. Therefore:

$$\hat{T}_{\mu\nu}{}^{\rho} = \frac{1}{6} g^{\rho\sigma} H_{\mu\nu\sigma} = \frac{1}{6} H_{\mu\nu}{}^{\rho} \neq 0 \quad \text{(for non-zero } H_{\mu\nu\rho}) \,. \tag{9}$$

Thus, the connection $\hat{\Gamma}_{\mu\nu}{}^{\rho}$ is **not torsion-free** (unless $H_{\mu\nu\rho} = 0$).

*2. Is $\hat{\Gamma}_{\mu\nu}{}^{\rho}$ metric compatible?*

A connection $\hat{\nabla}$ is metric compatible if and only if $\hat{\nabla}_{\mu} g_{\nu\rho} = 0$. Expanding the covariant derivative of the metric with respect to $\hat{\Gamma}_{\mu\nu}{}^{\rho}$:

$$\hat{\nabla}_{\mu} g_{\nu\rho} = \partial_{\mu} g_{\nu\rho} - \hat{\Gamma}_{\mu\nu}{}^{\lambda} g_{\lambda\rho} - \hat{\Gamma}_{\mu\rho}{}^{\lambda} g_{\nu\lambda} \,. \tag{10}$$

Substituting the definition (6) of $\hat{\Gamma}_{\mu\nu}{}^{\lambda}$:

$$\begin{aligned}
\hat{\nabla}_{\mu} g_{\nu\rho} &= \partial_{\mu} g_{\nu\rho} - \left(\Gamma_{\mu\nu}{}^{\lambda} + \frac{1}{12} g^{\lambda\sigma} H_{\mu\nu\sigma}\right) g_{\lambda\rho} - \left(\Gamma_{\mu\rho}{}^{\lambda} + \frac{1}{12} g^{\lambda\sigma} H_{\mu\rho\sigma}\right) g_{\nu\lambda} \\
&= \left(\partial_{\mu} g_{\nu\rho} - \Gamma_{\mu\nu}{}^{\lambda} g_{\lambda\rho} - \Gamma_{\mu\rho}{}^{\lambda} g_{\nu\lambda}\right) - \frac{1}{12} g^{\lambda\sigma} g_{\lambda\rho} H_{\mu\nu\sigma} - \frac{1}{12} g^{\lambda\sigma} g_{\nu\lambda} H_{\mu\rho\sigma} \\
&= \nabla_{\mu} g_{\nu\rho} - \frac{1}{12} \delta^{\sigma}_{\rho} H_{\mu\nu\sigma} - \frac{1}{12} \delta^{\sigma}_{\nu} H_{\mu\rho\sigma} \\
&= \nabla_{\mu} g_{\nu\rho} - \frac{1}{12} \left(H_{\mu\nu\rho} + H_{\mu\rho\nu}\right) \,.
\end{aligned} \tag{11}$$

Here $\nabla_\mu g_{\nu\rho} = 0$ because the Levi-Civita connection $\Gamma_{\mu\nu}{}^\lambda$ is metric compatible by definition. Moreover, because $H_{\mu\nu\rho}$ is totally antisymmetric, it is antisymmetric in its last two indices: $H_{\mu\rho\nu} = -H_{\mu\nu\rho}$, which implies $H_{\mu\nu\rho} + H_{\mu\rho\nu} = 0$. Hence:

$$\hat{\nabla}_\mu g_{\nu\rho} = 0\,. \tag{12}$$

Therefore, the connection $\hat{\Gamma}_{\mu\nu}{}^\rho$ **is metric compatible**.

b) **Equivalence of geodesics:**

An affinely parametrised curve $x^\mu(\lambda)$ with tangent vector $\dot{x}^\mu = \frac{dx^\mu}{d\lambda}$ is a geodesic (autoparallel curve) of the connection $\hat{\Gamma}_{\mu\nu}{}^\rho$ if it satisfies the geodesic equation:

$$\dot{x}^\mu \hat{\nabla}_\mu \dot{x}^\rho = \frac{d^2x^\rho}{d\lambda^2} + \hat{\Gamma}_{\mu\nu}{}^\rho \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,. \tag{13}$$

Substituting equation (6) into the geodesic equation gives:

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}{}^\rho \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,. \tag{14}$$

In the third term, the product $\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}$ is symmetric under the exchange of the dummy indices $\mu \leftrightarrow \nu$, whereas $H_{\mu\nu\sigma}$ is antisymmetric under $\mu \leftrightarrow \nu$ ($H_{\mu\nu\sigma} = -H_{\nu\mu\sigma}$). Contracting a symmetric tensor with an antisymmetric tensor yields zero identically:

$$H_{\mu\nu\sigma}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = \frac{1}{2}\left(H_{\mu\nu\sigma} + H_{\nu\mu\sigma}\right)\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,. \tag{15}$$

Consequently, the $H_{\mu\nu\sigma}$ contribution drops out completely, and the geodesic equation for $\hat{\Gamma}_{\mu\nu}{}^\rho$ reduces to

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}{}^\rho \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,, \tag{16}$$

which is the exact geodesic equation for the Levi-Civita connection $\Gamma_{\mu\nu}{}^\rho$. Thus, the geodesics of $\hat{\Gamma}_{\mu\nu}{}^\rho$ are identical to the geodesics of the Levi-Civita connection.

## Question 3. *(4 points)*

The Schwarzschild metric is given by

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2\,, \qquad f(r) = 1 - \frac{2GM}{r}\,. \tag{17}$$

a) **Singularities in the Schwarzschild metric and physical interpretation:**

*1. Locations of the singularities:*

Examining the metric components $g_{\mu\nu}$ and inverse metric components $g^{\mu\nu}$ in the Schwarzschild coordinates $(t, r, \theta, \phi)$, we see that pathologies occur at:

- $r = 2GM$ (the **Schwarzschild radius / event horizon**): here $f(r) = 0$, so $g_{tt} = 0$ and $g_{rr} = \left(1 - \frac{2GM}{r}\right)^{-1} \to \infty$ (while $g^{tt} \to -\infty$).
- $r = 0$ (the **central singularity**): here $f(r) \to -\infty$, so $g_{tt} \to \infty$, $g^{rr} \to -\infty$, and the two-sphere metric components $g_{\theta\theta}, g_{\phi\phi} \to 0$.
- ($\theta = 0, \pi$: here $g_{\phi\phi} = r^2 \sin^2\theta = 0$ and $g^{\phi\phi} \to \infty$, which is simply the standard coordinate singularity of spherical polar coordinates on $S^2$.)

*2. Distinguishing coordinate vs. physical singularities:*

Because metric components depend on the choice of coordinates, a divergence in $g_{\mu\nu}$ or $g^{\mu\nu}$ does not necessarily mean spacetime itself is singular. To test whether a singularity is a true physical (curvature) singularity or merely an artifact of bad coordinates, we examine coordinate-independent scalar invariants built from the Riemann tensor. Since the Schwarzschild metric is a vacuum solution ($R_{\mu\nu} = 0$ and $R = 0$), the relevant non-vanishing curvature invariant is the **Kretschmann scalar**:

$$K = R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6} \,. \tag{18}$$

- **At $r = 2GM$:** The Kretschmann scalar (along with all other curvature invariants) is completely finite:

$$R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}|_{r=2GM} = \frac{48G^2M^2}{(2GM)^6} = \frac{3}{4(GM)^4} < \infty \,. \tag{19}$$

  Therefore, $r = 2GM$ is only a **coordinate singularity**, resulting from the breakdown of the Schwarzschild coordinates $(t, r)$ at the horizon. Locally, the spacetime geometry is completely smooth at $r = 2GM$, and for a sufficiently supermassive black hole ($M$ large), the tidal forces at $r = 2GM$ are small enough that you can fall across $r = 2GM$ without feeling anything unusual at that moment.

- **At $r = 0$:** The Kretschmann scalar diverges:

$$\lim_{r\to 0} R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = \lim_{r\to 0} \frac{48G^2M^2}{r^6} = \infty \,. \tag{20}$$

  Because a scalar takes the same value in every coordinate system, this divergence cannot be removed by any coordinate transformation. Thus, $r = 0$ is a **true physical (curvature) singularity**.

*3. Which should you be genuinely concerned about, and why?*

- You must be genuinely concerned about the **physical curvature singularity at $r = 0$**, because as $r \to 0$ the spacetime curvature and gravitational tidal forces diverge to infinity, stretching you radially and compressing you transversely ("spaghettification") until you are destroyed.

- Furthermore, even though $r = 2GM$ is not a curvature singularity, you should **also** be concerned about crossing $r = 2GM$ because it is an **event horizon** (a causal boundary of no return). Once you cross into the interior region $r < 2GM$, the coordinate $r$ becomes timelike ($g_{rr} < 0$) and all future-directed timelike and null worldlines are forced toward decreasing $r$. You can never escape back to $r > 2GM$ or send any signal to your friends and family, and you are inevitably doomed to hit the lethal curvature singularity at $r = 0$ in a finite proper time ($\Delta\tau \leq \pi GM$).

b) **Schwarzschild metric in ingoing Eddington-Finkelstein coordinates:**

We are given the ingoing Eddington-Finkelstein coordinate

$$v = t + r + 2GM \ln\left|\frac{r}{2GM} - 1\right| . \tag{21}$$

Taking the differential $dv$ (valid for both $r > 2GM$ and $r < 2GM$ since $\frac{d}{dx}\ln|x| = \frac{1}{x}$):

$$\begin{aligned} dv &= dt + \left(1 + \frac{2GM}{\frac{r}{2GM} - 1} \cdot \frac{1}{2GM}\right) dr \\ &= dt + \left(1 + \frac{2GM}{r - 2GM}\right) dr \\ &= dt + \left(\frac{r}{r - 2GM}\right) dr \\ &= dt + \left(1 - \frac{2GM}{r}\right)^{-1} dr = dt + f(r)^{-1} dr \,. \end{aligned} \tag{22}$$

Rearranging for $dt$ and squaring gives:

$$dt = dv - f(r)^{-1} dr \,, \tag{23}$$

$$dt^2 = dv^2 - 2f(r)^{-1} dv\, dr + f(r)^{-2} dr^2 \,. \tag{24}$$

Substituting $dt^2$ into the Schwarzschild line element:

$$\begin{aligned} ds^2 &= -f(r)\left(dv^2 - 2f(r)^{-1} dv\, dr + f(r)^{-2} dr^2\right) + f(r)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \\ &= -f(r) dv^2 + 2dv\, dr - f(r)^{-1} dr^2 + f(r)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \\ &= -f(r) dv^2 + 2dv\, dr + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \,. \end{aligned} \tag{25}$$

Writing out $f(r) = 1 - \frac{2GM}{r}$ explicitly, the metric in ingoing Eddington-Finkelstein coordinates $(v, r, \theta, \phi)$ is:

$$\boxed{ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2dv\, dr + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \,.} \tag{26}$$

*How this supports the answer to part a):*

i. **Removal of the singularity at $r = 2GM$:** The singular $f(r)^{-1}dr^2$ term has cancelled identically against the cross-term from $dt^2$. At $r = 2GM$, $f(2GM) = 0$, and the metric (26) smoothly reduces to $ds^2 = 2dv\,dr + (2GM)^2(d\theta^2 + \sin^2\theta d\phi^2)$, which has completely finite, regular components. Moreover, the metric is invertible at $r = 2GM$ because $\det(g_{\mu\nu}) = -r^4\sin^2\theta \neq 0$ (the $(v, r)$ block $\begin{pmatrix} -f(r) & 1 \\ 1 & 0 \end{pmatrix}$ has determinant $-1$ everywhere). This explicitly demonstrates that the singularity at $r = 2GM$ in Schwarzschild coordinates was merely a **coordinate singularity**, and that an infalling observer crosses $r = 2GM$ smoothly along a regular worldline (e.g. at finite $v$).

ii. **Persistence of the physical singularity at $r = 0$:** In contrast, at $r = 0$ the metric component $g_{vv} = -\left(1 - \frac{2GM}{r}\right)$ still blows up to $+\infty$, consistent with $r = 0$ being a genuine **physical curvature singularity** that cannot be removed by changing coordinates.

iii. **Causal trapping inside the horizon ($r \leq 2GM$):** Radial null rays ($d\theta = d\phi = 0$, $ds^2 = 0$) satisfy

$$dv\left[-\left(1 - \frac{2GM}{r}\right)dv + 2dr\right] = 0\,, \tag{27}$$

giving $dv = 0$ (ingoing light rays) and $\frac{dr}{dv} = \frac{1}{2}\left(1 - \frac{2GM}{r}\right)$ (outgoing light rays). At $r = 2GM$, $\frac{dr}{dv} = 0$, so outgoing light is frozen at the horizon; for $r < 2GM$, $1 - \frac{2GM}{r} < 0$, so $\frac{dr}{dv} < 0$ for *all* future-directed null and timelike curves ($ds^2 \leq 0$ with $dv > 0$). Thus any observer who falls past $r = 2GM$ cannot escape and is forced into the singularity at $r = 0$.

## Question 4. *(5 points)*

The de Sitter metric is

$$ds^2 = -dt^2 + L^2\cosh^2(t/L)\left(d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta d\phi^2\right)\right)\,, \tag{28}$$

where $t \in (-\infty, \infty)$, $\chi \in (0, \pi)$, and $(\theta, \phi)$ are standard coordinates on $S^2$.

a) **de Sitter as a candidate cosmological solution and the scale factor:**

*Why it is a candidate cosmological solution for our universe:*

i. **Homogeneity and Isotropy (Cosmological Principle):** On large scales ($\gtrsim$ 100 Mpc), our universe is observed (via the Cosmic Microwave Background and large-scale galaxy surveys) to be spatially homogeneous and isotropic. In the metric (28), the spatial slices of constant cosmic time $t$ are three-spheres $S^3$ with round metric $d\Omega_3^2 = d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2)$. Because $S^3$ is a maximally symmetric 3-space (of constant positive curvature $k = +1$), the de Sitter metric

is a special case of the **Friedmann-Lemaître-Robertson-Walker (FLRW)** metric:

$$ds^2 = -dt^2 + a(t)^2 \left( d\chi^2 + \sin^2 \chi \left( d\theta^2 + \sin^2 \theta d\phi^2 \right) \right) . \tag{29}$$

ii. **Accelerated Expansion and Dark Energy / Inflation:** The de Sitter metric solves Einstein's equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ with a positive cosmological constant $\Lambda = \frac{3}{L^2} > 0$ (vacuum energy). For $t > 0$, the spatial sections expand ($\dot{a} > 0$) at an accelerating rate ($\ddot{a} = \frac{a}{L^2} > 0$), growing exponentially as $a(t) \approx \frac{L}{2} e^{t/L}$ for $t \gg L$. This describes two crucial eras of our physical universe: the early-universe epoch of **cosmic inflation**, and the present/future **dark-energy-dominated era** in which our universe's expansion is accelerating and asymptotically approaching de Sitter space.

*Scale factor:*

Comparing the metric (28) directly with the FLRW line element $ds^2 = -dt^2 + a(t)^2 d\Omega_3^2$, the scale factor is:

$$\boxed{a(t) = L \cosh\left(\frac{t}{L}\right)} \tag{30}$$

(or $a(t) = \cosh(t/L)$ if the curvature radius $L$ is absorbed into the spatial metric on $S^3$).

b) **Why conformally related metrics $g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu}$ share the same causal structure:**

The causal structure of a spacetime is defined by the light cones at every point—that is, by the partitioning of tangent vectors $V^\mu$ (and smooth curves $x^\mu(\lambda)$ with tangent $V^\mu = \frac{dx^\mu}{d\lambda}$) into timelike, null (lightlike), and spacelike categories.

Suppose $g_{\mu\nu}(x) = \Omega^2(x) \bar{g}_{\mu\nu}(x)$, where $\Omega(x)$ is a smooth, strictly non-vanishing function, so that $\Omega^2(x) > 0$ everywhere. For any tangent vector $V^\mu$, its norm squared in the two metrics is related by:

$$g_{\mu\nu} V^\mu V^\nu = \Omega^2 \bar{g}_{\mu\nu} V^\mu V^\nu \qquad \left(\text{equivalently, } ds^2 = \Omega^2 d\bar{s}^2\right) . \tag{31}$$

Because $\Omega^2 > 0$ is strictly positive, multiplication by $\Omega^2$ preserves the sign (and zeros) of the quadratic form at every point:

- **Timelike:** $g_{\mu\nu} V^\mu V^\nu < 0 \iff \bar{g}_{\mu\nu} V^\mu V^\nu < 0$,
- **Null (lightlike):** $g_{\mu\nu} V^\mu V^\nu = 0 \iff \bar{g}_{\mu\nu} V^\mu V^\nu = 0$,
- **Spacelike:** $g_{\mu\nu} V^\mu V^\nu > 0 \iff \bar{g}_{\mu\nu} V^\mu V^\nu > 0$.

Thus, the light cones (and hence all timelike, null, and spacelike curves, as well as causal future/past relations between events) of $g_{\mu\nu}$ and $\bar{g}_{\mu\nu}$ are identical. Consequently, $g_{\mu\nu}$ and $\bar{g}_{\mu\nu}$ have the exact same causal structure.

c) **Conformal coordinate transformation and Penrose diagram of de Sitter space:**

We perform the coordinate transformation from cosmic time $t \in (-\infty, \infty)$ to conformal time $\eta$:

$$\cosh \frac{t}{L} = \frac{1}{\cos \eta} \,. \tag{32}$$

Since $\cosh(t/L) \in [1, \infty)$ for $t \in (-\infty, \infty)$, the conformal time $\eta$ takes values in the finite open interval

$$\eta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \,, \tag{33}$$

with $t \to -\infty \implies \eta \to -\frac{\pi}{2}$, $t = 0 \implies \eta = 0$, and $t \to +\infty \implies \eta \to +\frac{\pi}{2}$.

Differentiating both sides of equation (32) gives:

$$\frac{1}{L} \sinh\left(\frac{t}{L}\right) dt = \frac{\sin \eta}{\cos^2 \eta} d\eta \,. \tag{34}$$

Using the hyperbolic and trigonometric identities $\sinh^2(t/L) = \cosh^2(t/L) - 1 = \frac{1}{\cos^2 \eta} - 1 = \frac{\sin^2 \eta}{\cos^2 \eta} = \tan^2 \eta$ (with $\mathrm{sgn}(t) = \mathrm{sgn}(\eta)$, so $\sinh(t/L) = \frac{\sin \eta}{\cos \eta}$), we obtain:

$$\frac{1}{L} \frac{\sin \eta}{\cos \eta} dt = \frac{\sin \eta}{\cos^2 \eta} d\eta \quad \implies \quad dt = \frac{L}{\cos \eta} d\eta \,. \tag{35}$$

Substituting $dt = \frac{L}{\cos \eta} d\eta$ and $\cosh(t/L) = \frac{1}{\cos \eta}$ into the de Sitter metric (28) yields:

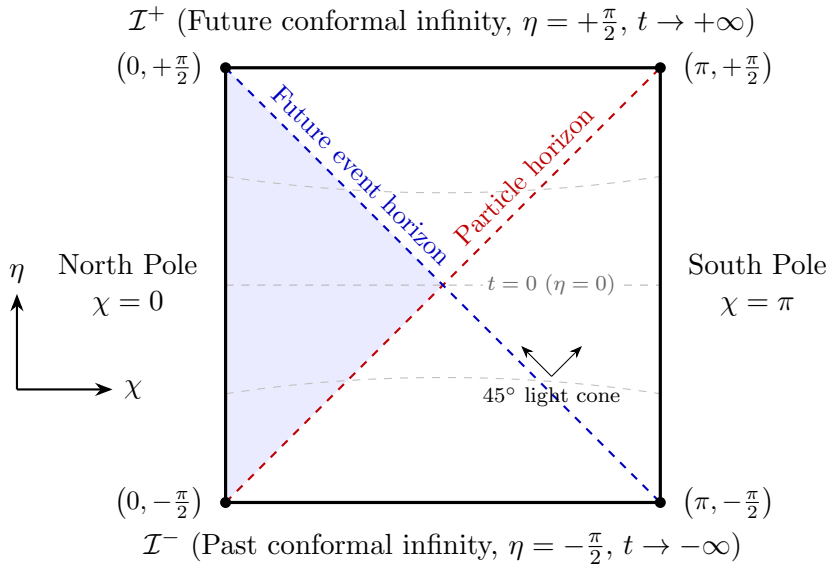
$$\begin{aligned} ds^2 &= -\frac{L^2}{\cos^2 \eta} d\eta^2 + \frac{L^2}{\cos^2 \eta} \left(d\chi^2 + \sin^2 \chi \left(d\theta^2 + \sin^2 \theta d\phi^2\right)\right) \\ &= \frac{L^2}{\cos^2 \eta} \left[-d\eta^2 + d\chi^2 + \sin^2 \chi \left(d\theta^2 + \sin^2 \theta d\phi^2\right)\right] \,. \end{aligned} \tag{36}$$

This is of the form $g_{\mu\nu} = \Omega^2(\eta) \bar{g}_{\mu\nu}$ with conformal factor $\Omega(\eta) = \frac{L}{\cos \eta}$ and unphysical metric

$$d\bar{s}^2 = -d\eta^2 + d\chi^2 + \sin^2 \chi \left(d\theta^2 + \sin^2 \theta d\phi^2\right) \,, \qquad \eta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \chi \in [0, \pi] \,. \tag{37}$$

In these coordinates, radial null rays ($d\theta = d\phi = 0, d\bar{s}^2 = 0$) obey $d\eta = \pm d\chi$, which are straight lines at $\pm 45^\circ$ in the $(\chi, \eta)$ plane.

**Penrose Diagram of de Sitter Spacetime:**




*Key features of the Penrose diagram:*

- The diagram is a **square** of width $\Delta\chi = \pi$ and height $\Delta\eta = \pi$.
- Each interior point $(\chi, \eta)$ with $\chi \in (0, \pi)$ represents an $S^2$ of conformal radius $\sin\chi$. The vertical lines $\chi = 0$ and $\chi = \pi$ are non-singular coordinate boundaries corresponding to the North and South poles of the spatial $S^3$ where the $S^2$ shrinks to a point.
- Past conformal infinity $\mathcal{I}^-$ ($\eta = -\frac{\pi}{2}, t \to -\infty$) and future conformal infinity $\mathcal{I}^+$ ($\eta = +\frac{\pi}{2}, t \to +\infty$) are **spacelike** horizontal boundaries. Consequently, any single observer (e.g. along $\chi = 0$) has both a cosmological **future event horizon** ($\eta = \frac{\pi}{2} - \chi$) and a **particle horizon** ($\eta = -\frac{\pi}{2} + \chi$).

## Question 5. *(6 points)*

The linearised Einstein equation for perturbations $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ ($|h_{\mu\nu}| \ll 1$) in terms of the trace-reversed perturbation $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}h\eta_{\mu\nu}$ (where $h = \eta^{\mu\nu}h_{\mu\nu}$) is:

$$-\frac{1}{2}\partial^\rho\partial_\rho\bar{h}_{\mu\nu} + \partial^\rho\partial_{(\mu}\bar{h}_{\nu)\rho} - \frac{1}{2}\eta_{\mu\nu}\partial^\rho\partial^\sigma\bar{h}_{\rho\sigma} = 8\pi G T_{\mu\nu}\,. \tag{38}$$

a) **Gauge choice and gravitational waves:**

Under an infinitesimal coordinate (gauge) transformation $x^\mu \mapsto x^\mu + \xi^\mu(x)$, the metric perturbation transforms as $h_{\mu\nu} \mapsto h_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu$, which induces the transformation on $\bar{h}_{\mu\nu}$:

$$\bar{h}_{\mu\nu} \mapsto \bar{h}_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu + \eta_{\mu\nu}\partial^\rho\xi_\rho \quad \Longrightarrow \quad \partial^\rho\bar{h}_{\mu\rho} \mapsto \partial^\rho\bar{h}_{\mu\rho} - \partial^\rho\partial_\rho\xi_\mu\,. \tag{39}$$

By choosing $\xi_\mu$ to satisfy the wave equation $\partial^\rho\partial_\rho\xi_\mu = \partial^\rho\bar{h}^{(\mathrm{old})}_{\mu\rho}$, we can always impose the **Lorenz (or de Donder / harmonic) gauge condition**:

$$\partial^\rho\bar{h}_{\mu\rho} = 0\,. \tag{40}$$

In this gauge, the second and third terms on the left-hand side of equation (38) vanish identically:

$$\partial^\rho\partial_{(\mu}\bar{h}_{\nu)\rho} = \frac{1}{2}\partial_\mu\left(\partial^\rho\bar{h}_{\nu\rho}\right) + \frac{1}{2}\partial_\nu\left(\partial^\rho\bar{h}_{\mu\rho}\right) = 0\,, \qquad \partial^\rho\partial^\sigma\bar{h}_{\rho\sigma} = 0\,. \tag{41}$$

The linearised Einstein equation (38) then simplifies to the decoupled wave equation:

$$\Box\bar{h}_{\mu\nu} \equiv \partial^\rho\partial_\rho\bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}\,. \tag{42}$$

In vacuum ($T_{\mu\nu} = 0$), this becomes the standard massless relativistic wave equation:

$$\Box\bar{h}_{\mu\nu} = \left(-\frac{\partial^2}{\partial t^2} + \nabla^2\right)\bar{h}_{\mu\nu} = 0 \tag{43}$$

(and after further fixing residual gauge freedom to the Transverse-Traceless gauge where $h = 0$, $\bar{h}_{\mu\nu} = h_{\mu\nu}$ satisfies $\Box h_{\mu\nu} = 0$). This equation admits plane-wave solutions $\bar{h}_{\mu\nu}(x) = \mathrm{Re}\left(\epsilon_{\mu\nu}e^{ik_\sigma x^\sigma}\right)$ with null wavevector $k^\sigma k_\sigma = -\omega^2 + |\mathbf{k}|^2 = 0$, demonstrating that perturbations of the gravitational field propagate through vacuum at the speed of light ($c = 1$) as **gravitational waves**.

b) **Proper distance between freely falling test particles:**

The metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, where in $(t, x, y, z)$ coordinates:

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+(t-z) & 0 & 0 \\ 0 & 0 & -h_+(t-z) & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{44}$$

*Step 1: Geodesic motion of the test particles.*

First, let us verify that freely falling test particles initially at rest ($\frac{dx^i}{d\tau} = 0$) remain at constant spatial coordinates $x^i$. The spatial geodesic equation is

$$\frac{d^2x^i}{d\tau^2} + \Gamma^i_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0\,. \tag{45}$$

For a particle initially at rest ($\frac{dx^i}{d\tau} = 0$, $\frac{dt}{d\tau} = 1$), this gives $\frac{d^2x^i}{d\tau^2} = -\Gamma^i_{00}$. To linear order in $h_{\mu\nu}$, since $h_{00} = 0$ and $h_{0i} = 0$:

$$\Gamma^i_{00} = \frac{1}{2}\eta^{ij}\left(2\partial_0 h_{j0} - \partial_j h_{00}\right) = 0\,. \tag{46}$$

Hence $\frac{d^2x^i}{d\tau^2} = 0$, so each freely falling test particle remains at its fixed spatial coordinate position $(0,0,0)$, $(L_x,0,0)$, or $(0,L_y,0)$ for all time $t$.

*Step 2: Proper distance along the x-axis.*

Both particles at $(0,0,0)$ and $(L_x,0,0)$ lie in the $z=0$ plane along the $x$-axis ($y=z=0$). At fixed time $t$ ($dt=dy=dz=0$), the line element along the $x$-axis at $z=0$ is:

$$ds^2 = g_{xx}(t,x,0,0)\,dx^2 = (\eta_{xx} + h_{xx}(t,z=0))\,dx^2 = (1+h_+(t))\,dx^2\,. \tag{47}$$

Integrating $ds = \sqrt{1+h_+(t)}\,dx$ from $x=0$ to $x=L_x$ and Taylor expanding to first order in the small perturbation $|h_+(t)| \ll 1$ using $\sqrt{1+\epsilon} \approx 1 + \frac{1}{2}\epsilon$:

$$\Delta s = \int_0^{L_x} \sqrt{1+h_+(t)}\,dx = L_x\sqrt{1+h_+(t)} \approx L_x\left(1+\frac{1}{2}h_+(t)\right)\,. \tag{48}$$

*Step 3: Proper distance along the y-axis.*

For the third test particle at $x^i = (0,L_y,0)$ (which also remains at fixed coordinates in the $z=0$ plane), we integrate along the $y$-axis from $(0,0,0)$ to $(0,L_y,0)$ at fixed time $t$ ($dt=dx=dz=0$). Here the line element is:

$$ds^2 = g_{yy}(t,0,y,0)\,dy^2 = (\eta_{yy} + h_{yy}(t,z=0))\,dy^2 = (1-h_+(t))\,dy^2\,. \tag{49}$$

Integrating $ds = \sqrt{1-h_+(t)}\,dy$ from $y=0$ to $y=L_y$ and expanding to first order in $h_+(t)$ gives:

$$\Delta s = \int_0^{L_y} \sqrt{1-h_+(t)}\,dy = L_y\sqrt{1-h_+(t)} \approx L_y\left(1-\frac{1}{2}h_+(t)\right)\,. \tag{50}$$

This exhibits the quadrupolar nature of gravitational waves: when the proper distance along the $x$-axis stretches by $+\frac{1}{2}L_xh_+(t)$, the proper distance along the perpendicular $y$-axis compresses by $-\frac{1}{2}L_yh_+(t)$, and vice versa half a wave period later.

c) **Binary coalescence, interferometer detection, and the LIGO waveform:**

*1. How a laser interferometer (such as LIGO) detects a gravitational wave:*

A gravitational-wave detector like LIGO is a large Michelson laser interferometer with two perpendicular arms along the $x$- and $y$-axes of equal unperturbed length $L_x = L_y = L$ (4 km for LIGO). A coherent laser beam at the origin $(0,0,0)$ is split by a beam splitter into two orthogonal beams that travel down the $x$- and $y$-arms, reflect off suspended test-mass mirrors at $(L,0,0)$ and $(0,L,0)$ (which are free to move horizontally as freely falling test particles), and recombine at the beam splitter.

When a gravitational wave such as that in part b) passes through the detector, one arm stretches while the perpendicular arm shrinks:

$$\Delta s_x(t) \approx L\left(1+\frac{1}{2}h_+(t)\right)\,, \qquad \Delta s_y(t) \approx L\left(1-\frac{1}{2}h_+(t)\right)\,. \tag{51}$$

This creates a time-dependent difference in proper arm lengths:

$$\Delta L(t) = \Delta s_x(t) - \Delta s_y(t) \approx L h_+(t) \quad \Longrightarrow \quad \frac{\Delta L(t)}{L} \approx h_+(t)\,. \tag{52}$$

Because light travels along null trajectories ($ds^2 = 0$, constant speed $c = 1$), the round-trip travel time of the laser light in the two arms differs by $\Delta t = 2\Delta L(t)/c \approx \frac{2L}{c} h_+(t)$, producing a relative phase shift $\Delta\phi(t) = \frac{4\pi}{\lambda_{\rm laser}} \Delta L(t) \approx \frac{4\pi L}{\lambda_{\rm laser}} h_+(t)$ between the two returning laser beams. This phase shift modulates the constructive/destructive interference intensity measured at the photodetector, directly tracking the gravitational wave strain $h_+(t)$.

*2. Qualitative evolution of a binary system and the three phases of the signal:*

A binary system of two massive compact objects (e.g. black holes or neutron stars of masses $m_1, m_2$) in a circular orbit possesses a time-varying mass quadrupole moment $Q_{ij}(t)$. According to general relativity, this accelerating quadrupole moment emits gravitational waves, which carry orbital energy and angular momentum away from the binary to infinity. Because the Newtonian orbital energy $E_{\rm orb} = -\frac{G m_1 m_2}{2R}$ is negative and becomes more negative as energy is lost ($\frac{dE_{\rm orb}}{dt} = -P_{\rm GW} < 0$), the orbital separation $R(t)$ must decrease.

The evolution and corresponding gravitational wave signal divide into three distinct regimes:

- **Inspiral:** Initially, the two bodies are well separated ($R \gg 2GM$) and orbit in a quasi-circular trajectory that gradually spirals inward as gravitational radiation drains orbital energy. By Kepler's third law, the orbital angular frequency is $\Omega_{\rm orb} = \sqrt{\frac{G(m_1+m_2)}{R^3}}$, so as the separation $R$ shrinks, the orbital frequency $\Omega_{\rm orb}$ and orbital velocity $v = \Omega_{\rm orb} R \propto R^{-1/2}$ *increase*. Because the mass quadrupole repeats every half-orbit, the gravitational wave frequency is twice the orbital frequency ($f_{\rm GW} = 2 f_{\rm orb} = \frac{\Omega_{\rm orb}}{\pi}$). Furthermore, the wave amplitude scales with the second time derivative of the quadrupole moment ($h \propto \ddot{Q} \propto M R^2 \Omega_{\rm orb}^2 \propto \frac{1}{R} \propto f_{\rm GW}^{2/3}$), and the radiated power scales strongly as $P_{\rm GW} \propto \langle \dddot{Q}^2 \rangle \propto R^4 \Omega_{\rm orb}^6 \propto R^{-5}$. Consequently, as the orbit shrinks, energy loss accelerates and **both the frequency and the amplitude of the gravitational wave increase simultaneously and ever more rapidly**, producing the characteristic upward-sweeping **"chirp"** waveform.
- **Merger:** As the orbital separation shrinks down to the innermost stable circular orbit (ISCO) and the scale of the horizons ($R \sim 2G(m_1 + m_2)$), the weak-field approximation breaks down and strong-field, highly non-linear general relativistic gravity dominates. The two bodies plunge rapidly together and coalesce into a single, highly deformed remnant object (e.g. a single black hole) moving at relativistic speeds ($v \sim c$). At this point of maximum spacetime curvature and

quadrupole acceleration, the gravitational wave signal reaches its **peak strain amplitude** and near-maximal frequency.

- **Ringdown:** Immediately following the merger, the newly formed remnant black hole is distorted away from equilibrium. In accordance with the black hole "no-hair" theorem, the perturbed horizon rapidly radiates away its asymmetries in the form of gravitational waves (quasinormal modes) until it settles into a stationary, axisymmetric Kerr black hole fully characterized by its final mass and spin. In the detector, this appears as an **exponentially damped high-frequency sinusoid**:

$$h(t) \sim Ae^{-(t-t_{\text{merger}})/\tau} \cos\left(\omega_{\text{QNM}} t + \phi_0\right) , \tag{53}$$

whose oscillation frequency $\omega_{\text{QNM}}$ and damping time $\tau$ are determined by the mass and spin of the final black hole, rapidly decaying to zero.

3. *Sketch of the gravitational wave signal observed by LIGO:*

